Investigación forense digital y respuesta ante incidentes sobre un endpoint Windows comprometido mediante análisis correlacionado de disco, registro, eventos y memoria

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Temas Avanzados de Seguridad Informática

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Introducción

“The hacker used Gnu to swap his special atrun file for the system's legitimate version. Five minutes later, the system hatched his egg, and he held the keys to my computer.”

Traducción:El hacker usó Gnu para intercambiar su archivo especial atrun por la versión legítima del sistema. Cinco minutos después, el sistema abrió su huevo, y él tenía las llaves de mi computadora.

(The Cucko Egg,Cliff Stoll,1989)

- **Análisis forense digital aplicado a ciberseguridad.**
- Endpoint **Windows comprometido**.
- Análisis de **disco, Registry, eventos y memoria**.
- Correlación de evidencias para **reconstruir el ataque**.

Capítulo I: Problemática

1.1 Contexto

1.2 Descripción de problema

1.3.1 Objetivo general

1.3.2 Objetivos específicos

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1.5.1 Alcance

1.5.2 Limitaciones

1.6 Cronograma de trabajo e hitos del proyecto

Problemática

1.1 Contexto

- Windows: **objetivo frecuente de ciberataques**.
- Ataques modernos: **fileless, ransomware y APT**.
- Evidencias distribuidas en múltiples artefactos.
- Necesidad de **correlacionar fuentes DFIR**.

1.2 Descripción del problema

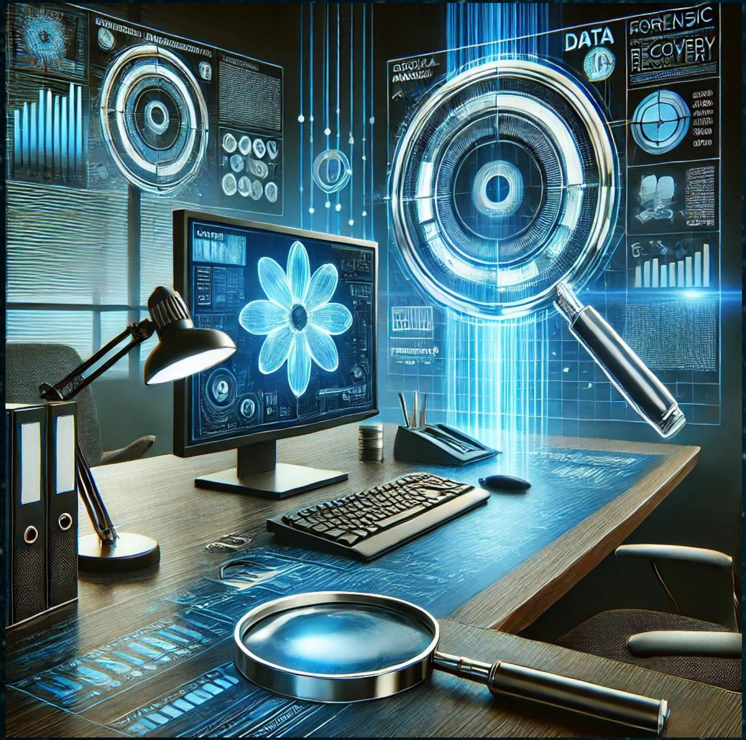
- Análisis forense frecuentemente **aislado por fuente**.
- Falta de una metodología **integral y estructurada**.
- Se pueden omitir **indicadores de compromiso**.
- Dificultad para reconstruir la **línea de tiempo del atacante**.

1.3.1 Objetivo general

Implementar un proceso de investigación forense digital y respuesta a incidentes para un endpoint Windows comprometido, correlacionando disco, Registry, eventos y memoria.

1.3.2 Objetivos específicos

- **Adquirir** evidencias con cadena de custodia e integridad.
- **Analizar** disco, Registry, Event Logs y memoria.
- **Correlacionar** evidencias y reconstruir la línea de tiempo.
- **Identificar** vector de entrada, persistencia e IoCs.
- **Elaborar** informe técnico con recomendaciones DFIR.



1.4 Justificación

•Fortalece competencias en **Digital Forensics & Incident Response**.
•Permite detectar **malware fileless y persistencia avanzada**.
•Integra evidencias **volátiles y no volátiles**.
•Aplica herramientas y metodologías **reconocidas en DFIR**.

1.5.1 Alcances

- **1 endpoint Windows 10** en laboratorio.
- Análisis de:
- **NTFS y artefactos de disco**
- **Registry**
- **Event Logs + Sysmon**
- **Memoria RAM**
- **Informe técnico** de investigación y respuesta.

1.5.2 Limitaciones

- **Sin sistemas ni datos de producción reales.**
- Escenario **controlado de laboratorio**.
- No contempla **red empresarial ni movimiento lateral**.
- Recursos de **hardware/almacenamiento** pueden limitar el análisis.

1.6 Cronograma de trabajo e Hitos

09 sep → 20 nov 2026

- **12 sep:** Primera entrega
- **30 sep:** Diseño cerrado
- **05 nov:** Avance de implementación
- **14 nov:** Documento completo
- **20 nov:** Entrega final

Distribución:

Brigitte→ Problemática
Milagros → Marco teórico
Stephany→ Arquitectura
Nardy→ Implementación
Marelly→ Pruebas y cierre

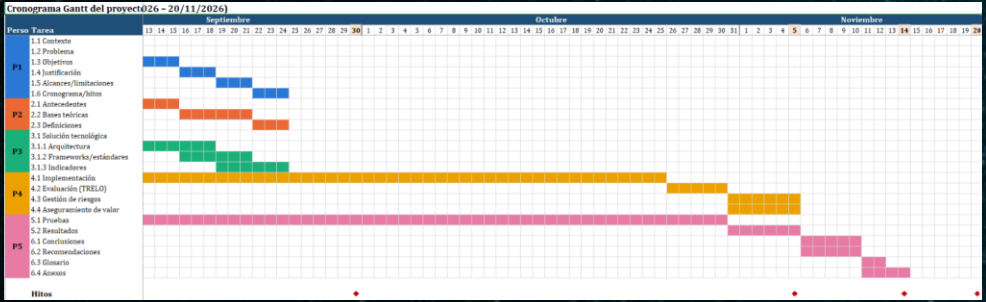


Diagrama de Gantt (Fig.1)

II

Marco teórico científico

III

DISEÑO METODOLÓGICO Y ARQUITECTURA

IV

IMPLEMENTACIÓN y GESTIÓN DEL PROYECTO

PRUEBAS, VALIDACIÓN Y RESULTADOS

VI

Conclusiones, recomendaciones,y anexos

LET'S HAVE A LOOK

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god. Despite being closer to the Sun than Venus, its temperatures aren't as terribly hot as that planet's. Its surface is quite similar to that of Earth's Moon, which means there are a lot of craters and plains. Speaking of craters, many of them were named after artists or authors who made significant contributions to their respective fields. Mercury takes a little more than 58 days to complete its rotation, so try to imagine how long days must be there!

THESE ARE TWO IDEAS

This is about Saturn

Saturn is a gas giant and has several rings. It's composed mostly of hydrogen and helium. It was named after the Roman god of wealth and agriculture

This is about Venus

Venus has a beautiful name and is the second planet from the Sun. Its atmosphere is extremely poisonous, primarily made up of carbon dioxide with clouds of sulfuric acid


A picture is worth a thousand words

THIS IS A TIMELINE

01

This is Venus

Venus has a beautiful name and is the second planet from the Sun. It's terribly hot, even hotter than Mercury

02

This is Earth

Earth is the third planet from the Sun and the only one that harbors life in the Solar System

03

This is Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust, which gives the planet its reddish cast

04

This is Jupiter

Jupiter is the fourth-brightest object in the night sky. It was named after the Roman god of the skies and lightning

05

This is Saturn

Saturn is a gas giant and has several rings. It's composed mostly of hydrogen and helium

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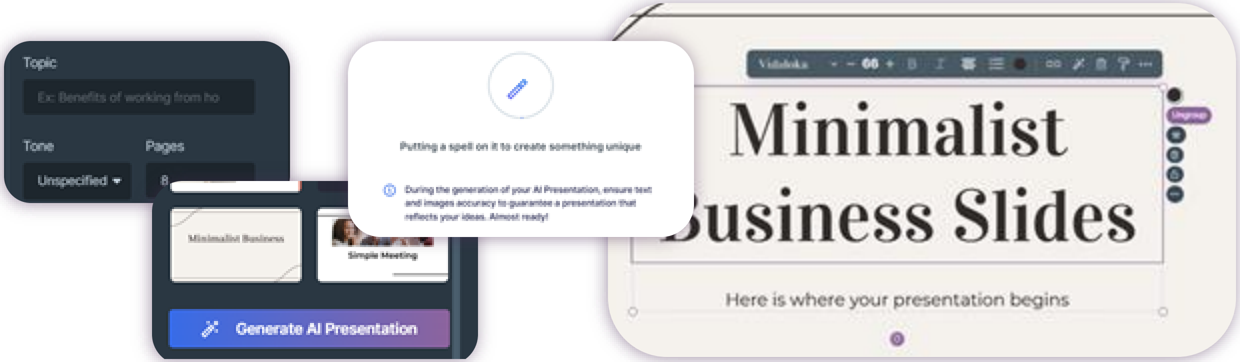

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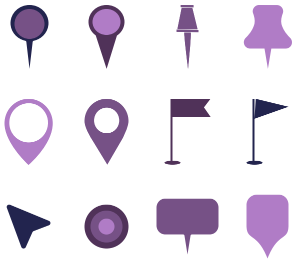

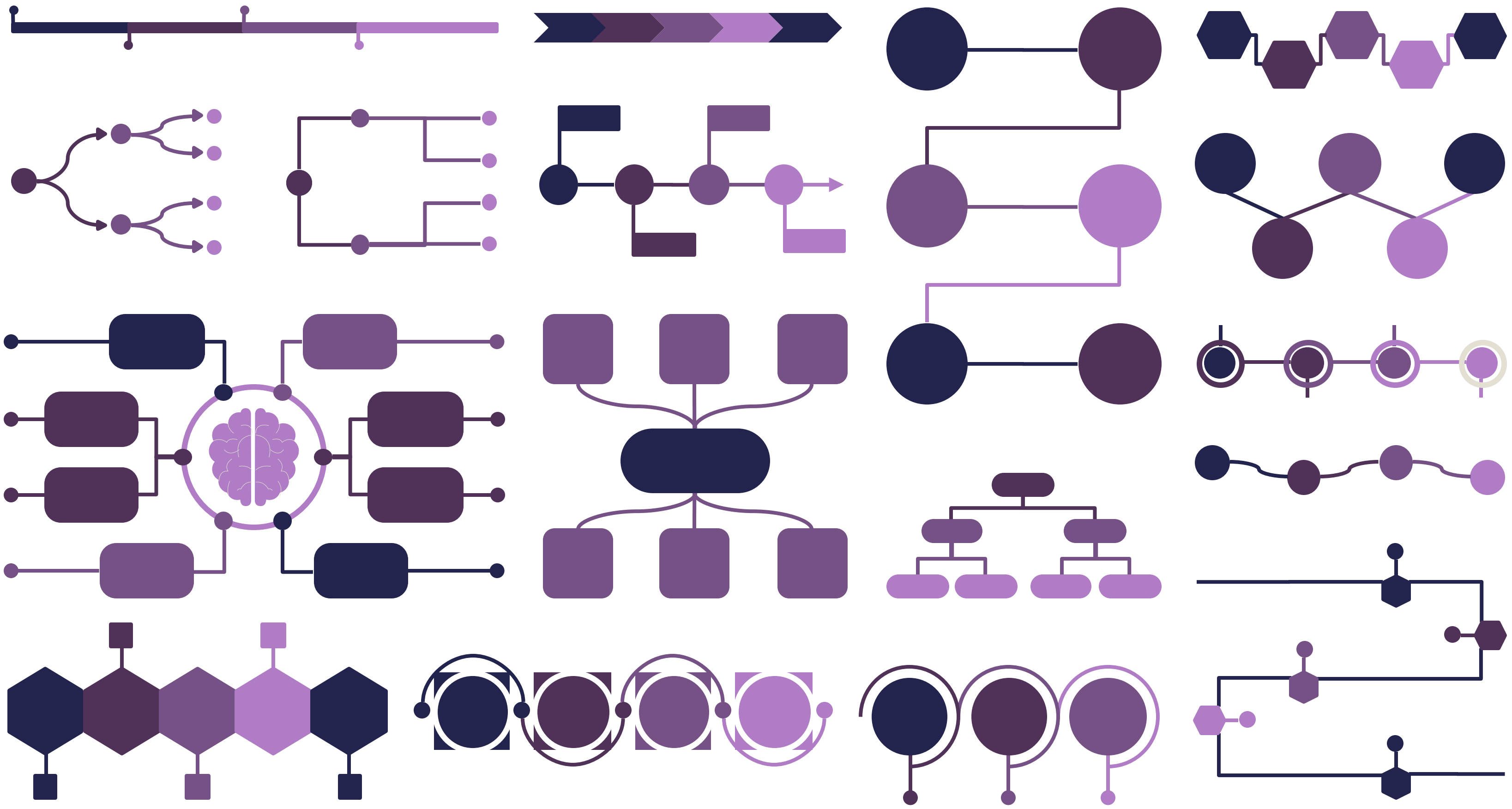

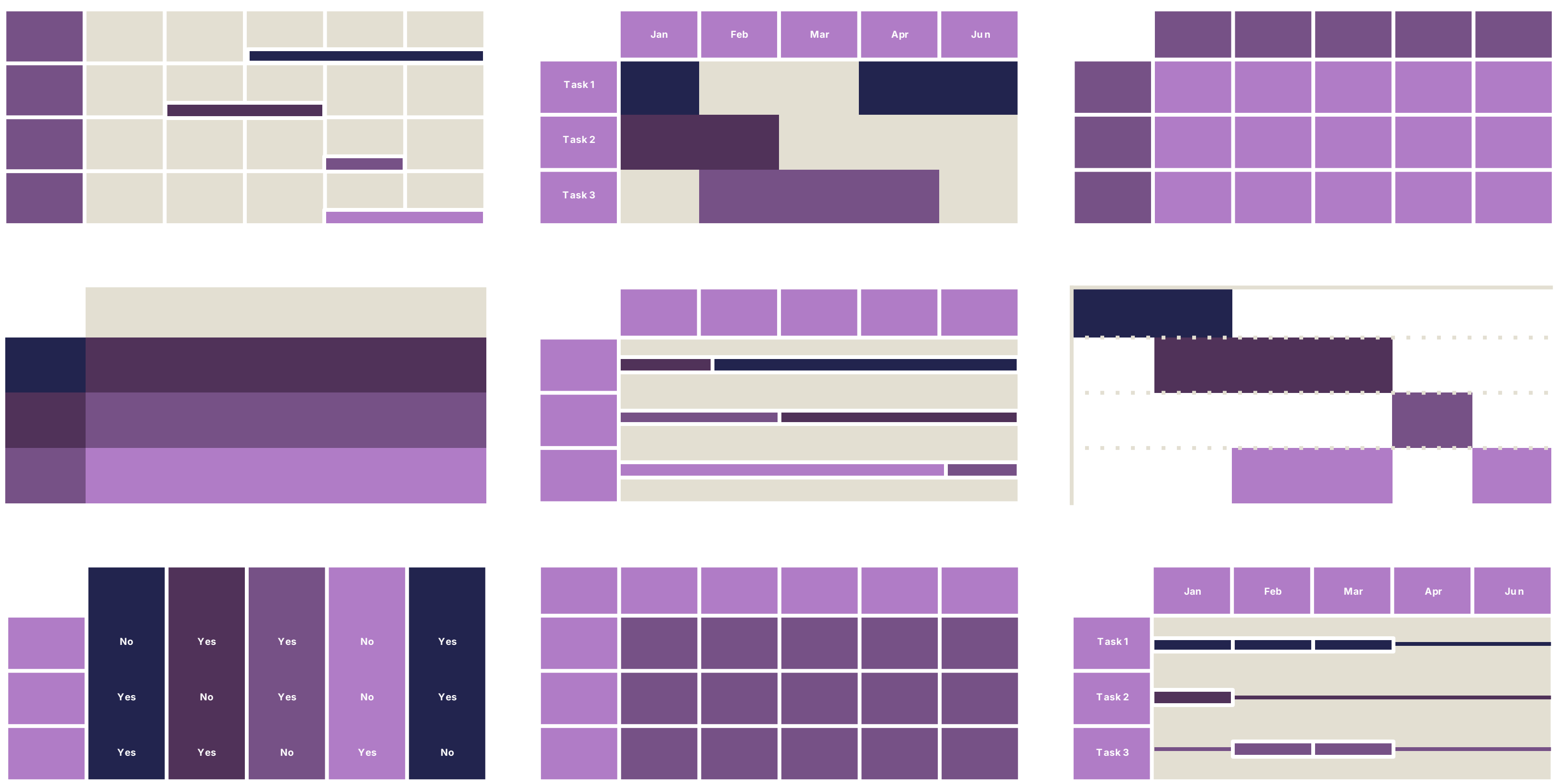

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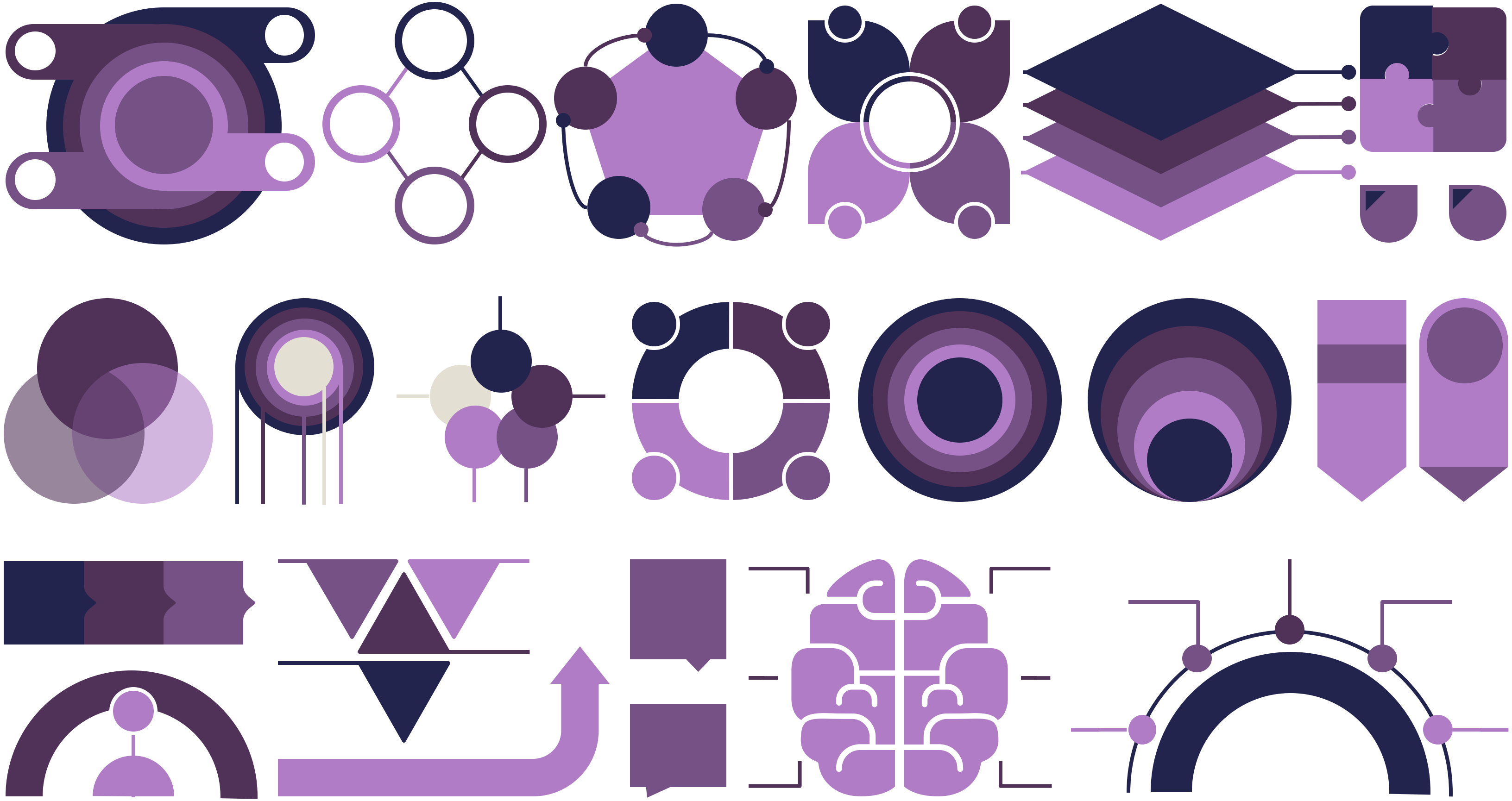


6/10
8/10
7/10
4/10

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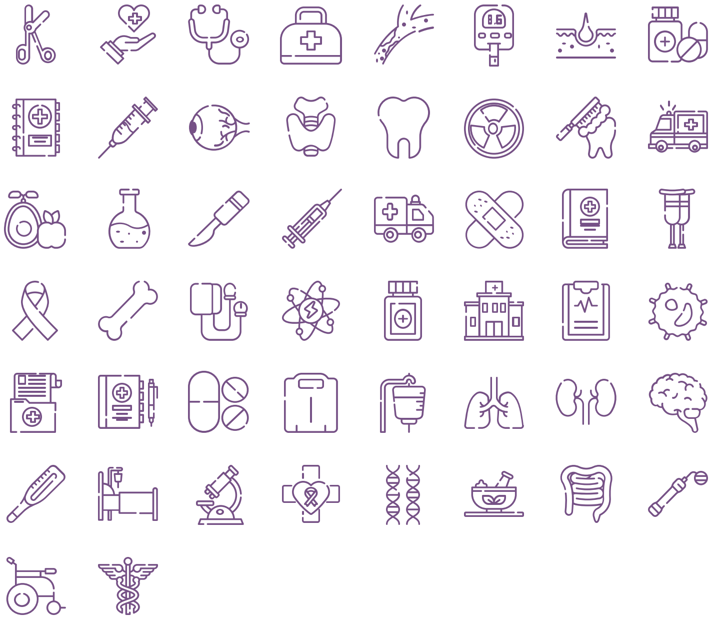
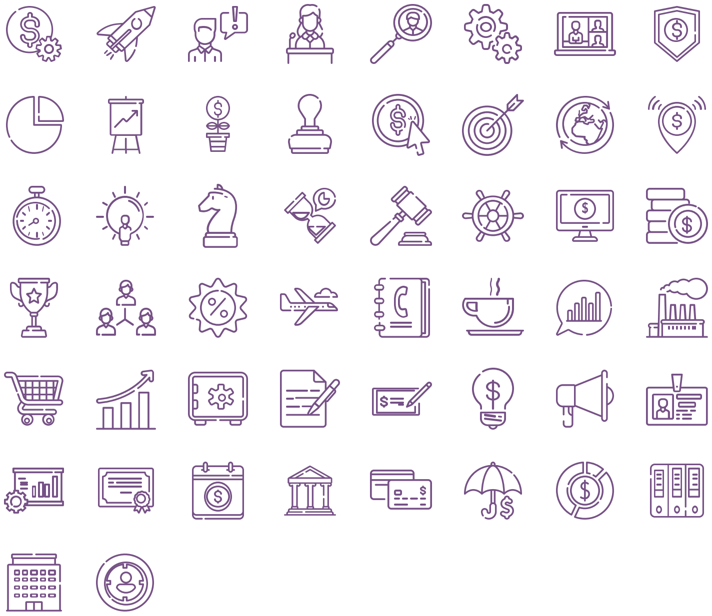
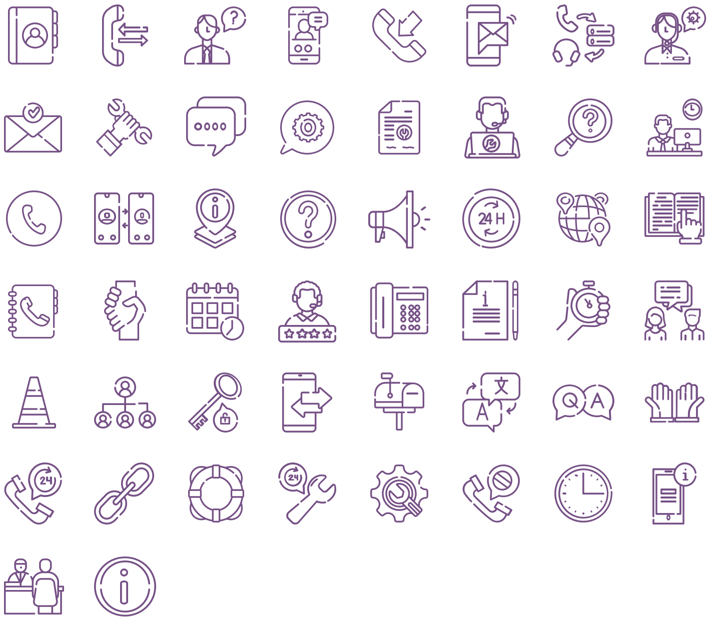


Business icons



Teamwork icons

Help & support icons



Avatar icons

Creative process icons

Performing arts icons

Nature icons

SEO & marketing icons

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Contexto

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Problemática

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Alcance

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Capítulo I: Problemática

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